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Characterisation of desorbing molecules by REMPI-TOF spectroscopy: the system H_2 (D_2)−V(1 0 0) + O

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Abstract

The internal state distribution (rotation, vibration) and the translational energy distribution of state selected molecules can be measured with the help of a REMPI (resonance enhanced multi-photon ionisation)-TOF (time-of-flight) spectrometer. We have performed measurements of this type on hydrogen and deuterium molecules desorbing from an oxygen covered V(1 0 0) surface. The rotational state population of the hydrogen and deuterium molecules in the vibrational ground state exhibits an overpopulation for $J<3$ and an underpopulation for $3<J<8$. In addition we observe a slight overpopulation for $J>9$ for desorbing deuterium. The calculation of the mean rotational energy $\langle E_{rot}\rangle$ leads in all cases to values significantly below the respective thermal equilibrium values (rotational cooling). For the vibrational state distribution we observe in all cases an overpopulation for molecules in the first excited vibrational state (vibrational heating). The translational energy distribution of desorbing deuterium strongly depends on the surface quality. On the flat, well annealed oxygen covered V(1 0 0) surface the mean translational energy of desorbing deuterium is hyperthermal, whereas the translational energy of molecules desorbing from a rough surface is closer to thermal.
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1. Introduction

The interaction of hydrogen with vanadium oxide has become a subject of considerable interest. It is well known that vanadium is able to absorb huge amounts of hydrogen and deuterium so that it is an interesting candidate for energy storage [1]. On the other hand vanadium is an important metal for heterogeneous catalysis [2]. Unfortunately, cleaning of the vanadium sample is very difficult. Therefore contaminated surfaces are mostly used for practical applications. There exist several papers dealing with the characterisation of contaminated vanadium surfaces [3,4] and adsorption on these surfaces [5,6].

We have focused on the desorption behaviour of hydrogen and deuterium from an oxygen covered V(1 0 0) surface. Using the REMPI-(resonance enhanced multi-photon ionisation) technique we have studied the rotational and vibrational state distribution of desorbing hydrogen molecules. With the time-of-flight (TOF) unit we can also obtain information on the translational energy of the state selected molecules. This should give us information on the particular potential energy

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